Pseudodeterministic Communication Complexity - medium

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1 Preliminaries

For $x, y \in \{0,1\}^n$, let

$$d_H(x,y) := \big|\{i \in [n] : x_i \neq y_i\}\big|$$

denote their Hamming distance.

Definition 1 (Gap Hamming Distance)**.** *The partial Boolean function*

$$\mathrm{GapHD}_n : \{0,1\}^n \times \{0,1\}^n \longrightarrow \{0,1,*\}$$

is defined by

$$\mathrm{GapHD}_n(x,y) := \begin{cases} 0, & d_H(x,y) \leq n/3, \\ 1, & d_H(x,y) \geq 2n/3, \\ *, & n/3 < d_H(x,y) < 2n/3. \end{cases}$$

Definition 2 (Pseudodeterministic communication complexity)**.** *For a total Boolean function $f : \{0,1\}^n \times \{0,1\}^n \to \{0,1\}$, let $\mathrm{R}_\varepsilon(f)$ be the minimum worst-case communication cost of a public-coin randomized protocol that outputs $f(x,y)$ with probability at least $1-\varepsilon$ on every input (x,y).*

A completion of a partial Boolean function $g : \{0,1\}^n \times \{0,1\}^n \to \{0,1,*\}$ is a total Boolean function f such that $f(x,y) = g(x,y)$ whenever $g(x,y) \neq *$. Define

$$\mathrm{PsR}_\varepsilon(g) := \min\big\{\mathrm{R}_\varepsilon(f) : f \text{ is a completion of } g\big\}.$$

2 Problem formulation

There exists an absolute constant $c > 0$ such that, for all sufficiently large n,

$$\mathrm{PsR}_{1/3}(\mathrm{GapHD}_n) \geq cn.$$

Equivalently, every completion $f : \{0,1\}^n \times \{0,1\}^n \to \{0,1\}$ of GapHD_n satisfies

$$\mathrm{R}_{1/3}(f) \geq cn.$$